OnePrompt-All: One Click for Same-Concept Multi-Instance Segmentation

Training-free point-conditioned multi-instance grounding with SAM 3

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Abstract

Can one click on an object be used to segment the other visually matching instances in the same image? We study this question with OnePrompt-All, a training-free extension of SAM 3. The clicked point conditions the grounding pathway while SAM 3's object queries continue to search the full image. Candidate masks are filtered by area and mask NMS, with optional DINOv2 similarity reranking. We evaluate **2,220 images** from **232 classes**, using five valid clicks per image (**11,100 trials**). OnePrompt-All obtains **F1 0.667** and **mIoU 0.601**. Performance still varies with click position.

Problem & Contributions

Research question

SAM 3 can use a point to segment one selected object, while text or exemplars can retrieve matching instances. We ask whether one click on an example object can retrieve the other matching instances in the same image.

Point prompt: click → selected object
Concept prompt: text / exemplar → matching instances
OnePrompt-All: click → matching instances

Contributions

1. **Formulation.** Same-concept multi-instance segmentation from one positive click, without class text or fine-tuning.
2. **Point-conditioned grounding.** Inject the clicked location into SAM 3's grounding pathway while retaining full-image object queries.
3. **Candidate-set construction.** Refine global candidates with area filtering and mask NMS; DINOv2 similarity reranking is optional.
4. **Evaluation.** Test 2,220 images from 232 classes with 11,100 click trials, including click-position robustness.

Method

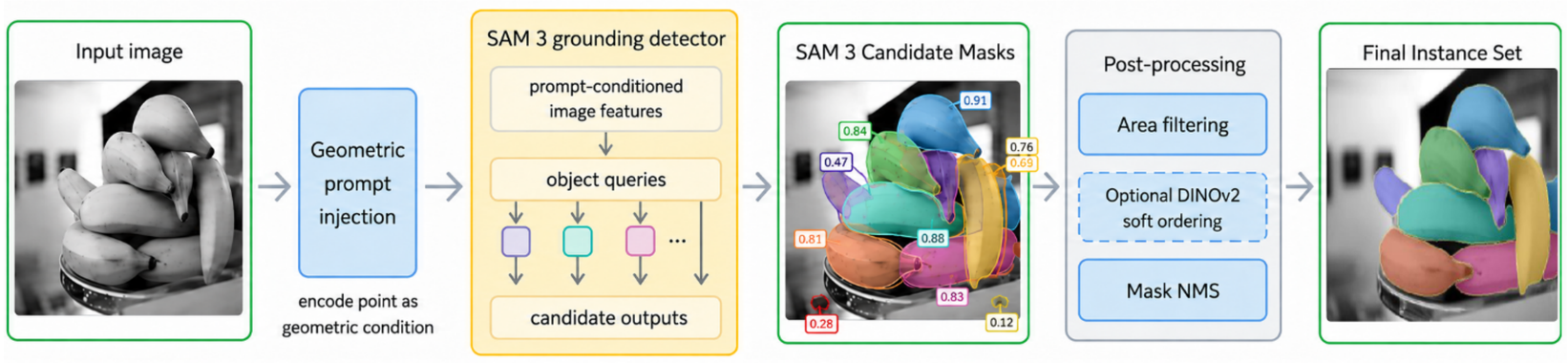
Point-conditioned global grounding

The clicked point is used as a geometric condition inside SAM 3's grounding pathway. The object queries are kept active over the full image, so the model can return multiple candidate masks instead of only the clicked object.

1. Point condition
Encode the clicked location and inject it into the grounding representation.

2. Global proposals
Keep SAM 3 object queries active across the image to generate multiple candidate masks.

3. Post-processing
Remove small masks, apply mask NMS, and optionally rerank candidates using DINOv2 similarity.



Test-time input: image + one positive click. No category text or fine-tuning.

Results

2,220	11,100	0.667	0.601	0.687	0.712
images	click trials	F1	mIoU	Precision	Recall
232 classes	5 clicks / image	precision-recall balance	mean mask overlap	candidate correctness	target-instance coverage

Evaluation: each run uses the image and one positive click; ground truth is used only to sample valid clicks and compute the metrics.

Click robustness

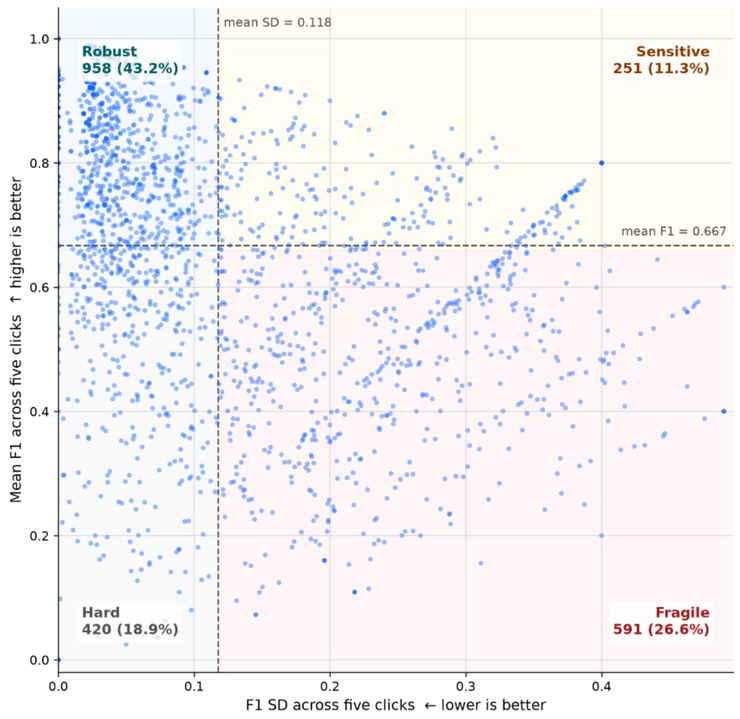


Fig. 2. Each dot is one image evaluated with five valid clicks. Higher mean F1 is better; lower F1 SD indicates greater click stability. The top-left region is therefore preferred.

Web demo

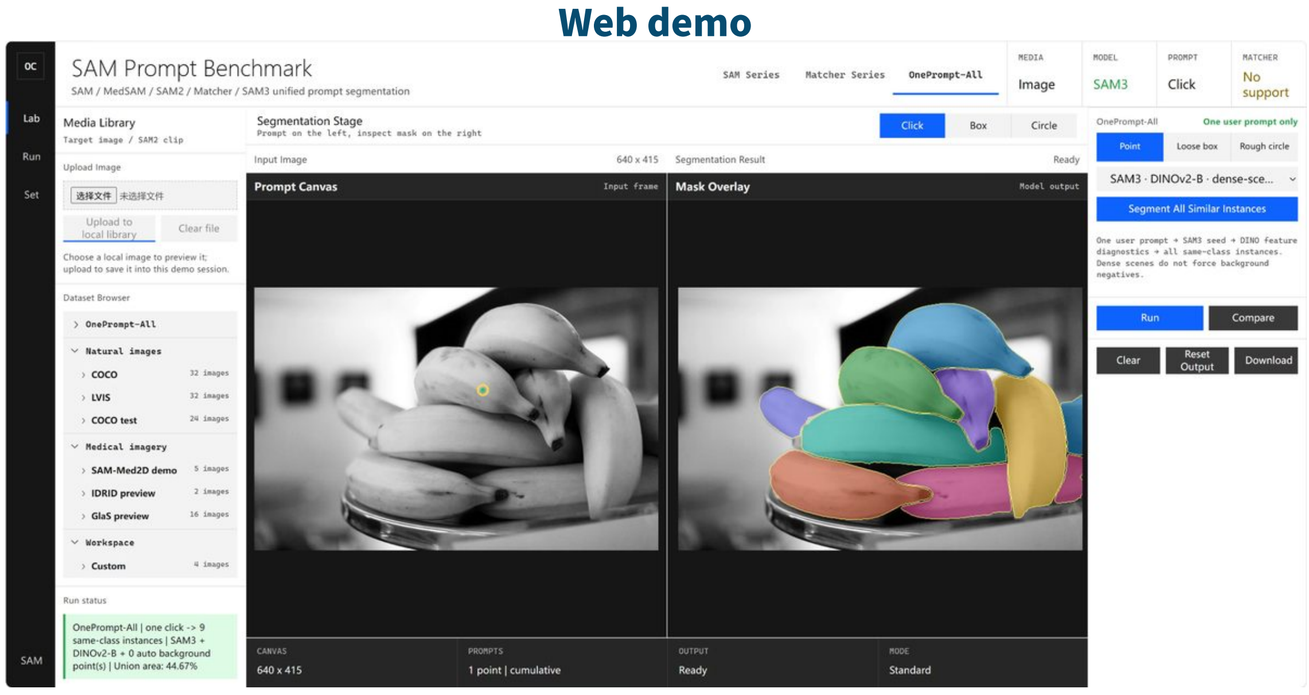


Fig. 3. The browser demo lets the user click one object and inspect all returned same-concept masks in the same interface.

Potential use. Repeated-object annotation, dataset curation, and interactive mask inspection.
Observed error pattern. Recall is slightly higher than precision (0.712 vs. 0.687), indicating some extra candidate masks.

Conclusion & Limitations

Conclusion

A single positive click can produce a multi-instance response from SAM 3 without category text or fine-tuning. On the integrated benchmark, OnePrompt-All reaches **F1 0.667** and **mIoU 0.601**.

Limitations & next step

Performance still changes with click position, and extra candidate masks reduce precision. Next, we will calibrate candidate confidence and compare point prompting directly with SAM 3 text- and exemplar-based prompting.

References

Carion et al., *SAM 3: Segment Anything with Concepts*, 2025.
Oquab et al., *DINOv2: Learning Robust Visual Features without Supervision*, 2023.

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